

Learn from Others and Be Yourself in Heterogeneous Federated Learning

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<https://github.com/WenkeHuang/FCCL>

Abstract

Federated learning has emerged as an important distributed learning paradigm, which normally involves collaborative updating with others and local updating on private data. However, heterogeneity problem and catastrophic forgetting bring distinctive challenges. First, due to non-i.i.d (identically and independently distributed) data and heterogeneous architectures, models suffer performance degradation on other domains and communication barrier with participants models. Second, in local updating, model is separately optimized on private data, which is prone to overfit current data distribution and forgets previously acquired knowledge, resulting in catastrophic forgetting. In this work, we propose FCCL (Federated Cross-Correlation and Continual Learning). For heterogeneity problem, FCCL leverages unlabeled public data for communication and construct cross-correlation matrix to learn a generalizable representation under domain shift. Meanwhile, for catastrophic forgetting, FCCL utilizes knowledge distillation in local updating, providing inter and intra domain information without leaking privacy. Empirical results on various image classification tasks demonstrate the effectiveness of our method and the efficiency of modules.

1. Introduction

Deep learning algorithms have achieved remarkable progress, owing to the availability of large-scale data [8, 51, 69]. However, in the real world, data are commonly dispersed over different participants (e.g., mobile devices, organizations). Due to growing privacy concerns and strict data protection regulations [84], participants cannot integrate data together to train a model. Driven by such realistic issues, federated learning [33, 34, 58, 59, 89] provides a privacy-preserving paradigm, where participants collaboratively

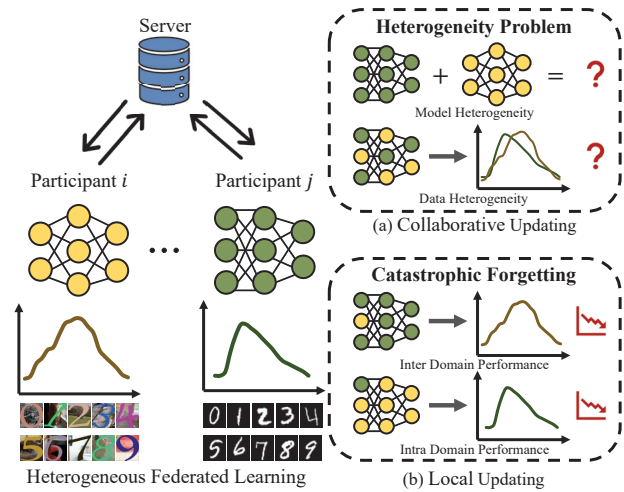


Figure 1. **Problem illustration of heterogeneous federated learning.** (a) In collaborative updating, how to handle communication problem of heterogeneous models and learn a generalizable representation under heterogeneous data (domain shift)? (b) In local updating, how to alleviate catastrophic forgetting to present stable and satisfactory performance in both inter and intra domains?

atively learn a model without leaking private data. It has been an active and challenging research topic and shows promising results in real-world setting [17, 19, 29, 52, 54].

Along with its pilot progress, researches on federated learning are baffled by some key challenges [30, 42]. An inevitable and practical challenge is heterogeneity problem. On the one hand, distributed data might be non-i.i.d (identically and independently distributed), leading to **data heterogeneity** [30, 39, 95]. A myriad of methods [43, 46, 73, 77] incorporate extra proximal terms to handle the data in **label distribution skew** (prior probability shift) [30], neglecting the fact that there exists **domain shift** (same label, different features) [60, 64, 66]. In particular, private model suffers severe performance degradation on other domains with no-

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ticeably different distribution. As a result, learning a generalizable representation under domain shift is technically challenging. On the other hand, due to different design criteria, distinct hardware capabilities [20, 86] and intellectual property rights [56], participants require to customize models, which poses a practical challenge: **model heterogeneity**. Preceding methods are developed under the assumption that local models share parameters or gradients, which cannot work on heterogeneous models. In order to solve this problem, a main stream of subsequent effort leverages knowledge transfer through labeled data [38, 74], shared model [48, 72, 92] or group operation [21, 50]. But these methods have different limitations. Specifically, labeled data require server to collect data with similar distributions to private data, which causes costly human efforts and needs special domain expertise. For shared model, it raises computational cost and necessitates additional model structure in participant side. Group operation leverages unlabeled public data to measure distribution divergence. However, these methods mainly focus on label distribution skew and consider the performance on one domain. Simultaneously considering data and model heterogeneity, an essential issue has long been overlooked: (a) *How to learn a generalizable representation in heterogeneous federated learning?*

Besides heterogeneity problem, another impediment for federated learning stems from its paradigm. Generally, federated learning could be viewed as a two-step cyclic process: *collaborative updating* and *local updating* [58, 89]. In *collaborative updating*, participants learn from others. In *local updating*, model is optimized on private data, which is prone to overfit current knowledge and forget previous knowledge, resulting in **catastrophic forgetting** [57]. To tackle this challenge, one type of methods typically performs fine-tuning for several rounds [38, 50, 58, 74, 88]. However, carefully configuring hyper-parameters to achieve satisfactory performance is time-consuming and cannot tackle this problem systematically. Current popular solutions [41, 43, 73, 77] focus on calculating parameter stiffness to regulate models, which can not explicitly depict the degree of effect from different participants. Consequently, a natural question arises: (b) *How to balance multiple knowledge to reduce catastrophic forgetting?* We further explain heterogeneity problem and catastrophic forgetting in Fig. 1.

For the heterogeneity problem, we take inspiration from the self-supervised learning [5, 6, 11, 13, 18, 25, 49, 91, 94]. In particular, self-supervised learning aims to learn a generalizable representation through rich and diverse data for downstream tasks and unseen classes. Intuitively, we expect that the models would present similar logits output for the same classes in different domains. This motivates us to leverage unlabeled public data for *Federated Cross-Correlation Learning*, which is diverse and easy to obtain. Specifically, we try to maximize the similarity between log-

its output and minimize the redundancy within logits output on unlabeled public data. Through correlating same dimensions and decorrelating different dimensions on logits output, models would learn class invariance and encourage the diversity of different classes. Thus, our method handles the communication problem in heterogeneous models and learns a generalizable representation under domain shift.

To handle catastrophic forgetting, we develop *Federated Continual Learning* via knowledge distillation [2, 24] in local updating to continually learn from inter and intra domains. To avoid forgetting inter domain information in local updating stage, we propose to distill the knowledge of intra-domain (local) model learned in previous rounds, where it captured the inter domain information after communication with other participants. In addition, for intra domain forgetting problem, we leverage the initially pretrained local model (without knowledge learned from others) to constrain the later local updating for each participant. Therefore, balancing knowledge through distillation with these two models is reasonable to handle the catastrophic forgetting.

In this work, we propose a novel federated learning method, dubbed *FCCL* (**F**ederated **C**ross-**C**orrelation and **C**ontinual **L**earning). The overview of *FCCL* is illustrated in Fig. 2. In a nutshell, our contributions are three-fold:

- We formulate a simple and effective method for heterogeneous federated learning. Through leveraging unlabeled public data and adopting self-supervised learning, heterogeneous models achieve communication and learn a generalizable representation.
- We explore to alleviate catastrophic forgetting in federated learning. Through inter and intra domain knowledge distillation with updated and pretrained models, it balances knowledge from others and itself.
- We conduct extensive experiments on two image classification tasks (*e.g.*, *Digits* [27, 37, 62, 68] and *Office-Home* [82]) with unlabeled public data [35, 69, 87]. *FCCL* achieves superior inter and intra domain performance over related methods. Ablation study on core module validates its efficacy and indispensability.

2. Related Work

Federated with Data Heterogeneity. A pioneering work proposed the currently most widely used algorithm, *FedAVG* [58]. But it suffers performance deterioration on non-i.i.d data (data heterogeneity). Shortly after, a large body methods [12, 41, 43, 73, 77] research on non-i.i.d data. These methods mainly focus on label distribution skew, where non-i.i.d data [30] are formed by partitioning existing data based on label space with limited domain shift. However, when private data sampled from different data

domains, these works do not consider inter domain performance but only focus on learning an internal model. Latest researches have studied related problems of unsupervised domain adaptation for target domain [45, 65] and domain generalization on unseen domains [52]. However, collecting data in the target domain can be time-consuming and impractical. Meanwhile, considering the performance on unknown domains is an idealistic setting. For more realistic settings, participants are probably more interested in the performance on other domains, which could directly improve economic benefits. In this work, we focus on improving inter domain performance under domain shift.

Federated with Model Heterogeneity. With the demand for unique models, federated learning with model heterogeneity has been an active area of research. *FedMD* [38], *CRONUS* [4] and *CFD* [71] operate on labeled public data (with similar distribution) via knowledge distillation [2, 24]. Therefore, these approaches heavily rely on the quality of labeled public data, which may not always be available on the server. Latest works (e.g., *FedDF* [50], *FedKT* [40] and *FEDGEN* [75]) have proven the feasibility to do distillation on unlabeled public data or synthetic data. However, these methods leverage unlabeled public data to reach semantic information consistency through various measuring metrics [9, 36], which are not suitable to learn a generalizable representation and thus lead to a bad inter domain performance. Another direction is introducing shared extra model such as *FML* [72] and *LG-FEDAVG* [48]. However, these techniques may not be applicable when considering the additional computing overhead and expensive communication cost. In this paper, based on unlabeled public data, we correlate same dimensions and decorrelate different dimensions to learn a generalizable representation in heterogeneous federated learning.

Self-Supervised Learning. Self-supervised Learning has emerged as a powerful method for learning useful representation without supervision from labels, largely reducing the performance gap between supervised models on various downstream vision tasks. Many related methods rely on contrastive learning (e.g., *SimCLR* [5], *MoCO* [7, 22]), which contrast positive pairs against negative pairs and minimizes difference between positive pairs for avoiding collapsing solutions [79, 90]. Recently, another line of works (e.g., *BYOL* [15], *SimSiam* [6]) employs asymmetry of the learning update (stop-gradient operation) to avoid trivial solutions. Besides, some methods (e.g., *W-MSE* [3], *Barlow Twins* [91]) investigate the possibility of feature decorrelation based on *Cholesky Decomposition* [83] and *Information Bottleneck* [80]. There are several works that consider federated learning with self-supervised learning (e.g., *FURL* [93], *MOON* [41]). They focus on the unsupervised learning setting and label distribution skew with model homogeneity respectively. The key difference between *FCCL*

and above self-supervised learning methods is that ours is designed for federated setting rather centralized setting. Inspired by self-supervised learning, *FCCL* constructs the comparison between different models in federated learning.

Catastrophic Forgetting. Catastrophic forgetting has been an essential problem in continual learning when models continuously learn from a stream data, with the goal of gradually extending acquired knowledge and using it for future learning [14, 57]. The challenge lies in the continuously changing class distributions of each task [63, 81]. Existing continual learning works on tackling catastrophic forgetting can be broadly divided into three branches [10]: replay methods [1, 67], regularization-based methods [32, 47, 53, 85] and parameter isolation methods [55, 61, 70]. As for federated learning, data are distributed rather than sequential like continual learning. But these differences aside, both continual learning and federated learning share a common challenge - how to balance the knowledge from different data distribution. Unlike continual learning methods, we focus on alleviate catastrophic forgetting in distributed data rather than time series data. In particular, we expect to balance and boost both inter and intra domain performance.

3. Method

Problem Setup and Notations. Following the standard federated learning setup, there are K participants (indexed by i). Each participant has a local model θ_i and private data $D_i = \{(X_i, Y_i) | X_i \in \mathbb{R}^{N_i \times D}, Y_i \in \mathbb{R}^{N_i \times C}\}$, where N_i denotes the number of private data, D represents input size and C is defined as the number of classes for classification. Meanwhile, the private data distribution is denoted as $P_i(X, Y)$ and rewritten as $P_i(X|Y)P_i(Y)$. Furthermore, in heterogeneous federated learning, **data heterogeneity** and **model heterogeneity** are defined as following:

- **Data heterogeneity:** $P_i(X|Y) \neq P_j(X|Y)$. There exists domain shift among private data, i.e., conditional distribution $P(X|Y)$ of private data vary across participants even if $P(Y)$ is shared. Specifically, same label Y has distinctive feature X in different domains.
- **Model heterogeneity:** $\text{Shape}(\theta_i) \neq \text{Shape}(\theta_j)$. Participants customize models independently, i.e., for classification task, the selected backbones (e.g., *ResNet* [23], *EfficientNet* [78] and *MobileNet* [26]) are different with differential classifier models.

We leverage unlabeled public data $D_0 = \{X_0 | X_0 \in \mathbb{R}^{N_0 \times D}\}$ to realize communication. The public data are relatively easy to access in real scenarios, e.g., existing datasets [8, 51, 69] and web images [44]. The goal for i^{th} participant is to reach communication and learn a model θ_i with generalizable representation. In addition, considering

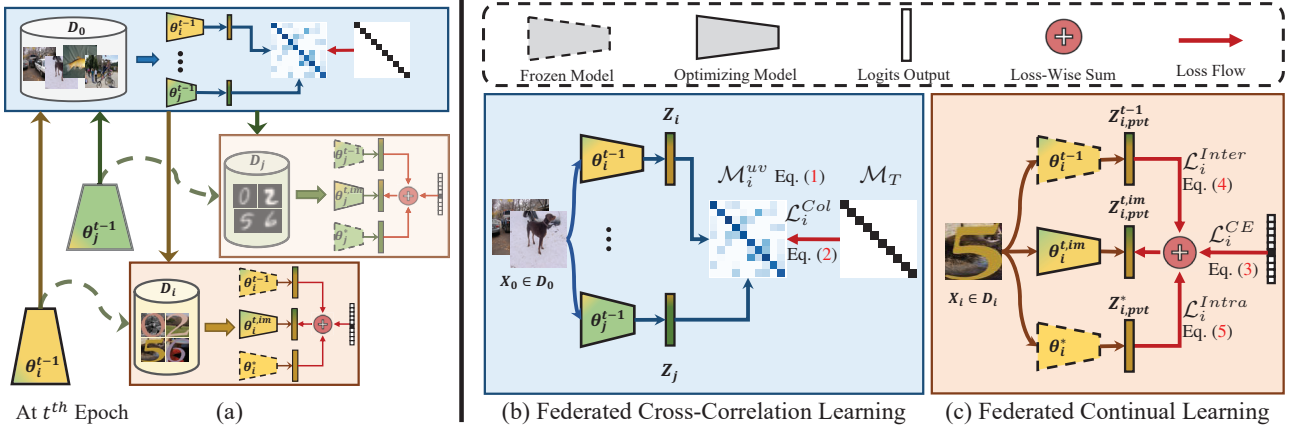


Figure 2. **Illustration of FCCL.** (a) Simplified schematization of our method that solves heterogeneity problem and catastrophic forgetting via *Federated Cross-Correlation Learning* and *Federated Continual Learning*. (b) *Federated Cross-Correlation Learning* § 3.1: Construct cross-correlation matrix $\mathcal{M}_i$ to target matrix, $\mathcal{M}_T = 2 \times \text{eye}(C) - \text{ones}(C)$, where on-diagonal is 1, off-diagonal is -1 . (c) *Federated Continual Learning* § 3.2: Distillation with updated and pretrained models offers inter and intra domain knowledge without privacy leaking. The gradient color proportion reflects the degree of influence by other participants. Best viewed in color. Zoom in for details.

catastrophic problem, θ_k is required to present both higher and stabler inter and intra domain performance.

Overview of Framework. The framework of our method is illustrated in Fig. 2. Specifically, in collaborative updating, we measure cross-correlation matrix between logits output on unlabeled public data to make similarity and reduce redundancy. Meanwhile, in local updating, ours continually balances multi domains information through knowledge distillation. Next, we will first describe *Federated Cross-Correlation Learning* § 3.1. Then we introduce *Federated Continual Learning* § 3.2.

3.1. Federated Cross-Correlation Learning

Motivation of Dimension-Level Operation. Motivated by the success of self-supervised learning via *Information Bottleneck* [80, 91], a generalizable representation should be as informative as possible about image, while being as invariant as possible to the specific domain distortions that are applied to this sample. In our work, domain shift results in distinctive feature X for the same label Y in different domains. Therefore, the distribution of logits output along the batch dimension on different domains is not identical. Moreover, different dimensions of logits output are corresponding to distinct classes. Thus, we need to encourage the invariance of same dimensions and the diversity of different dimensions. Private data carries specific domain information and is under privacy protection, which is not suitable and feasible to do self-supervised learning. Therefore, we leverage the unlabeled public data, which are normally generated and collected from multi domains and is easy to obtain. We optimize private models through requiring logits output invariant to domain distortion and decorrelating different dimensions of logits output on unlabeled public data.

Construction of Cross-Correlation Matrix. Specifically, we get the logits output for i^{th} participant: $Z_i = f(\theta_i, X_0) \in \mathbb{R}^{N_0 \times C}$. For i^{th} and j^{th} participant, the logits output on unlabeled public data is Z_i and Z_j . Notably, considering the computing burden on the server side, we calculate average logits output: $\bar{Z} = \frac{1}{K} \sum_i Z_i$. Then, we compute cross-correlation matrix, $\mathcal{M}_i$ for i^{th} participant with average logits output as:

$$\mathcal{M}_i^{uv} \triangleq \frac{\sum_b \|Z_i^{b,u}\| \|\bar{Z}^{b,v}\|}{\sqrt{\sum_b \|Z_i^{b,u}\|^2} \sqrt{\sum_b \|\bar{Z}^{b,v}\|^2}}. \quad (1)$$

where b indexes batch samples, u, v index the dimension of logits output and $\|\cdot\|$ is the normalization operation along the batch dimension. $\mathcal{M}_i$ is a square matrix with size of output dimensionality, C and values comprised between -1 (i.e., dissimilarity) and 1 (i.e., similarity). Then, collaborative loss for i^{th} participant is defined as:

$$\mathcal{L}_i^{Col} \triangleq \sum_u (1 - \mathcal{M}_i^{uu})^2 + \lambda_{Col} \sum_u \sum_{v \neq u} (1 + \mathcal{M}_i^{uv})^2, \quad (2)$$

where λ_{Col} is a positive constant trading off the importance of the first and second terms of loss. Naturally, when on-diagonal terms of the cross-correlation matrix take the value $+1$, it encourages the logits output from different participants to be similar; when off-diagonal terms of the cross-correlation matrix take value -1 , it encourages the diversity of logits output, since different dimensions of these logits output will be uncorrelated to each other.

Comparison with Analogous Methods. *FedMD* [38] relies on minimizing mean square error on annotated data. *FedDF* [50] reaches logits output distribution consistency on unlabeled public data. However, in our work, we expect

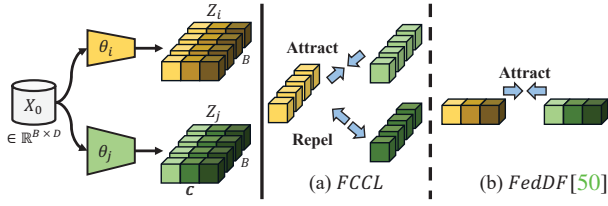


Figure 3. **Conceptual comparison.** The unlabeled public data X_0 with batch size B and input size D are fed into different models. The logits output has C dimensions. (a) *FCCL* learns invariance in same dimensions and decorrelates pairs of different dimensions on the batch-wise normalized logits output in Eq. (1). (b) *FedDF* [50] calculates the distribution divergence where instance-wise normalized logits output is compared inside a batch.

to achieve correlation of same dimensions but decorrelation of different dimensions on unlabeled public data. Besides, we do operation along the batch dimension, which means that we view unlabeled public data as ensemble rather than individual sample. It is advantageous to eliminate anomalous sample disturbance. We further illustrate the conceptual comparison between *FCCL* and *FedDF* in Fig. 3.

3.2. Federated Continual Learning

Typical Supervision Loss. For local updating in federated learning, current methods [38, 50, 58, 74] typically cast this process as a supervised classification problem. Specifically, at t^{th} communication round, after the collaborative updating, the i^{th} private model is defined as $\theta_i^{t,im}$. Then, optimize $\theta_i^{t,im}$ on private data $D_i(X_i, Y_i)$ for fixed epochs. Given the logits output $Z_{i,pvt}^{t,im} = f(\theta_i^{t,im}, X_i)$ for private data X_i w.r.t its ground truth label Y_i , the cross-entropy loss is optimized with softmax:

$$\mathcal{L}_i^{CE} = -\mathbf{1}_{Y_i} \log(\text{softmax}(Z_{i,pvt}^{t,im})), \quad (3)$$

where $\mathbf{1}_{Y_i}$ denotes the one-hot encoding of Y_i and $\text{softmax}(Z_{i,pvt}^{t,im}) = \frac{\exp(Z_{i,pvt}^{t,im})}{\sum_{c'=1}^C \exp(Z_{i,pvt}^{t,im,c'})}$. Such training objective design would suffer catastrophic forgetting mainly due to the following two limitations: **1)** In local updating, without supervision from other participants, models easily overfit current data distribution and present poor inter domain performance. **2)** Besides, it only penalizes the prediction independently with prior probabilities, which provides limited and hard intra domain information [24].

Dual-Domain Knowledge Distillation Loss. In this work, we develop a federated continual learning method to address both **1)** and **2)** through regularizing the objective from model-wise aspect. Specifically, at the end of $t-1^{th}$ round, the updated model, θ_i^{t-1} involves the knowledge learned from other participants. We calculate the logits output on private data: $Z_{i,pvt}^{t-1} = f(\theta_i^{t-1}, X_i)$. The inter

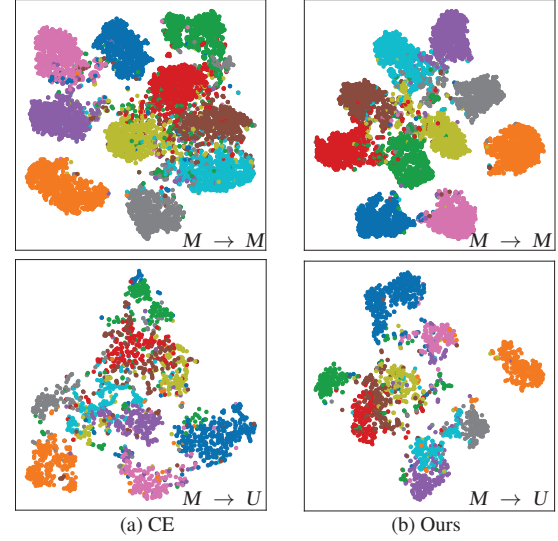


Figure 4. **Visualization of features** learned with (left) typical supervision loss (*i.e.*, cross-entropy loss, $\mathcal{L}^{CE}$ in Eq. (3) and (right) optimization objective based on dual-domain knowledge distillation (*i.e.*, $\mathcal{L}^{Dual}$ in Eq. (6)) on intra (top) and inter (bottom) domain. M and U represent *MNIST* and *USPS* respectively. Features are colored based on class labels.

domain knowledge distillation loss is defined as:

$$\mathcal{L}_i^{Inter} = \sigma(Z_{i,pvt}^{t-1}) \log \frac{\sigma(Z_{i,pvt}^{t,im})}{\sigma(Z_{i,pvt}^{t,im})}, \quad (4)$$

where σ denote softmax function. As Eq. (4), the purpose is to continually learn from others while preserving privacy, so as to guarantee inter domain performance and handle catastrophic forgetting in federated learning. Besides, for the i^{th} participant, it is feasible to pretrain a model, θ_i^* on private data. We measure the logits output on private data: $Z_{i,pvt}^* = f(\theta_i^*, X_i)$. The intra domain knowledge distillation loss can be given as :

$$\mathcal{L}_i^{Intra} = \sigma(Z_{i,pvt}^*) \log \frac{\sigma(Z_{i,pvt}^*)}{\sigma(Z_{i,pvt}^{t,im})}. \quad (5)$$

Knowledge distillation with pretrained model provides soft and rich intra domain information. Further, it cooperates with the former typical supervision loss (*i.e.*, cross-entropy loss) in Eq. (3) to provide soft and hard intra domain information to ensure intra domain performance. To some extent, above two models (*i.e.* updated model θ_i^{t-1} and pretrained model θ_i^*) respectively represent inter and intra ‘teacher’ model. Through knowledge distillation, balancing knowledge from others and itself simultaneously boosts both inter and intra domain performance. The dual-domain knowledge distillation is calculated by

$$\mathcal{L}_i^{Dual} = \mathcal{L}_i^{Inter} + \mathcal{L}_i^{Intra}. \quad (6)$$

Algorithm 1: The FCCL Framework

Input: Communication rounds T , local epochs E , participants number K , unlabeled public data (X_0) , i^{th} private data (X_i, Y_i) and pretrained model θ_i^* , hyper-parameter $\lambda_{Col}, \lambda_{Loc}$

for $t = 1, 2, \dots, T$ **do**

for $i = 1, 2, \dots, K$ **do**

 Calculate logits output: $Z_i = f(\theta_i^{t-1}, X_0)$

 Average logits output: $\bar{Z} = \frac{1}{K} \sum_i Z_i$

for $i = 1, 2, \dots, K$ **in parallel do**

$\theta_i^{t,im} \leftarrow$ **Federated Cross-Correlation Learning** $(Z_i, \bar{Z}, \theta_i^{t-1})$

$\theta_i^t \leftarrow$ **Federate Continual Learning** $(\theta_i^*, \theta_i^{t-1}, \theta_i^{t,im})$

return θ_i^T

Federated Cross-Correlation Learning $(Z_i, \bar{Z}, \theta_i^{t-1})$

$\mathcal{M}_i \leftarrow (Z_i, \bar{Z})$ by Eq. (1)

$\mathcal{L}_i^{Col} \leftarrow (\mathcal{M}_i, \lambda_{Col})$ through Eq. (2)

$\theta_i^{t,im} \leftarrow \theta_i^{t-1} - \eta \nabla \mathcal{L}_i^{Col}$

return $\theta_i^{t,im}$ to i^{th} participant

Federated Continual Learning $(\theta_i^*, \theta_i^{t-1}, \theta_i^{t,im})$:

for $e = 1, 2, \dots, E$ **do**

$Z_{i,pvt}^{t,im} = f(\theta_i^{t,im}, X_i)$

$\mathcal{L}_i^{CE} \leftarrow \text{CE}(Z_{i,pvt}^{t,im}, Y_i)$ in Eq. (3)

$\mathcal{L}_i^{Inter} \leftarrow \text{KL}(Z_{i,pvt}^{t,im}, f(\theta_i^{t-1}, X_i))$ in Eq. (4)

$\mathcal{L}_i^{Intra} \leftarrow \text{KL}(Z_{i,pvt}^{t,im}, f(\theta_i^*, X_i))$ in Eq. (5)

$\mathcal{L}_i^{Dual} = \mathcal{L}_i^{Inter} + \mathcal{L}_i^{Intra}$

$\mathcal{L}_i^{Loc} = \mathcal{L}_i^{CE} + \lambda_{Loc} \mathcal{L}_i^{Dual}$

$\theta_i^{t,im} \leftarrow \theta_i^{t,im} - \eta \nabla \mathcal{L}_i^{Loc}$

$\theta_i^t \leftarrow \theta_i^{t,im}$

return θ_i^t to i^{th} participant

The typical supervision loss in Eq. (3) and dual-domain knowledge distillation loss in Eq. (6) are complementary to each other. The former requires models to learn a discriminative representation that is meaningful for classification tasks, while the latter helps to regularize the model with soft and rich information in both intra and inter domain. Thus, the overall training target is:

$$\mathcal{L}_i^{Loc} = \mathcal{L}_i^{CE} + \lambda_{Loc} \mathcal{L}_i^{Dual}, \quad (7)$$

where $\lambda_{Loc} > 0$ is a coefficient. As shown in Fig. 4, the features learned by $\mathcal{L}_i^{Dual}$ is more compact and separated in both intra and inter domain by enjoying the advantage of both typical supervision loss and dual-domain knowledge distillation loss, models show better discriminative features, producing promising intra and inter domain performance.

3.3. Discussion and Limitation

We describe FCCL in Alg. 1. FCCL constructs cross-correlation matrix with the average logits output. Therefore, FCCL is applicable when there are a large size of partici-

pants in federated learning, attributed to that the computation complexity for server side is $\mathcal{O}(K)$. Besides, *Federated Cross-Correlation Learning* does operation on logits output regardless of the specific model structure. Thus, when participants share same model structure (model homogeneity), FCCL is still capable. Assuming that there is no data heterogeneity among distributed data, the first term of $\mathcal{L}_i^{Col}$ in Eq. (1) would be close to zero, but the second term still disassociates different dimensions on logits output. On this basis, FCCL is model agnostic method and able to handle different degree of domain shift. However, we also note limitation on the requirement of task consistency. For multi-task setting, logits output may not only have distinct dimensions, but also contain different meanings for same dimension. This limitation is also shared by related methods [38, 50, 74, 91].

4. Experiments

Data and Model. We extensively evaluate our method on two classification tasks (e.g., *Digits* [27, 37, 62, 68] and *Office-Home* [82]) with three public data (e.g., *Cifar-100* [35], *ImageNet* [69] and *Fashion-MNIST* [87]). Specifically, *Digits* task includes four domains (i.e., *MNIST* (M), *USPS* (U), *SVHN* (SV) and *SYN* (SY)) with 10 categories. The *Office-Home* task also have four domains (i.e., *Art* (A), *Clipart* (C), *Product* (P) and *Real World* (R)). Note that for both tasks, data acquired from different domains present domain shift (**data heterogeneity**).

For these two classification tasks, participants customize models that can be differ from differentiated backbones and classifiers (**model heterogeneity**). For experiments, we set the model as *ResNet* [23], *EfficientNet* [78], *MobileNet* [26] and *GoogLeNet* [76] for these four domains.

Comparison Methods. We compare our method, FCCL with state-of-the-art approaches including *FedDF* [50], *FML* [72], *FedMD* [38], *RCFL* [16] and *FedMatch* [28]. We also compare *SOLO*, where participant trains a model on private data without federated learning. Since specific experimental settings are not totally consistent, we retain key features of methods for comparison.

Evaluation Metrics. We report the standard metrics to measure the quality of methods: accuracy, which is defined as the number of samples that are paired divided by the number of samples. Specifically, for evaluation intra and inter domain performance, we define as following:

$$\mathcal{A}_i^{Intra} = \frac{\sum (\arg\text{Max}(f(\theta_i, X_j^{Test})) == Y_j^{Test})}{|D_i^{Test}|}, \quad (8)$$

$$\mathcal{A}_i^{Inter} = \sum_{j \neq i} \frac{\sum (\arg\text{Max}(f(\theta_i, X_j^{Test})) == Y_j^{Test})}{(K-1) \times |D_j^{Test}|}. \quad (9)$$

As for the method overall performance evaluation, we

Methods	<i>Digits</i>					<i>Office-Home</i>				
	$M \rightarrow$	$U \rightarrow$	$SV \rightarrow$	$SY \rightarrow$	AVG	$A \rightarrow$	$C \rightarrow$	$P \rightarrow$	$R \rightarrow$	AVG
<i>SOLO</i>	15.29	13.91	39.24	34.30	25.68	18.89	19.36	21.97	21.02	20.31
<i>FedMD</i> [38]	8.97	12.61	40.89	43.03	26.38	16.85	23.13	28.78	25.01	23.44
<i>FML</i> [72]	17.11	16.00	45.19	46.26	31.14	18.97	24.41	29.75	24.91	24.51
<i>RCFL</i> [16]	10.21	16.10	<u>48.85</u>	37.96	28.28	15.16	22.01	27.98	23.95	22.28
<i>FedDF</i> [50]	13.23	19.29	45.25	43.95	30.43	17.38	21.76	25.17	22.97	21.82
<i>FedMatch</i> [28]	9.22	14.76	46.28	36.05	26.58	19.05	25.24	28.73	24.35	24.34
<i>FCCL</i>	<u>20.74</u>	<u>20.60</u>	44.68	<u>48.02</u>	33.51	<u>25.55</u>	<u>26.41</u>	<u>30.14</u>	<u>29.41</u>	27.88

Table 1. **Comparison of inter domain performance with state-of-the-art methods.** $M \rightarrow$ means that private data is *MNIST* and respective model is tested on other domains in Eq. (9). AVG denotes average accuracy calculated from each domain. (The best average accuracy is marked in bold. The best entries in each domain are underlined. These notes are the same to others.)

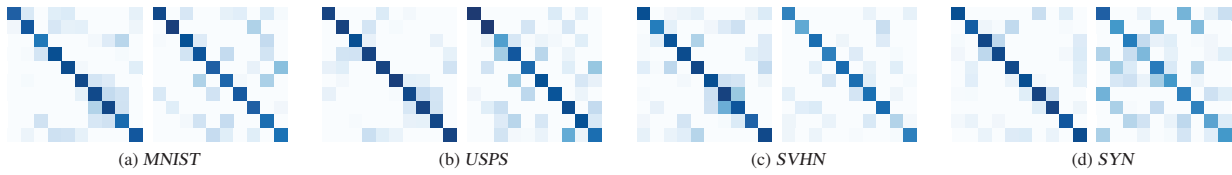


Figure 5. **Cross-correlation matrix visualization** for different domains on *Digits* task with *Cifar-100*. We visualize the cross-correlation matrix (Eq. (1)) with other models on public data (left) and private data (right) respectively. The left and right figure in each subfigure represent the cross-correlation matrix with other models on public data (*i.e.*, *Cifar-100*) and private data respectively. The matrix is 10×10 . The darker the color, the closer the $\mathcal{M}_i^{uv}$ (Eq. (1)) is to 1.

adopt the average accuracy as metric. Besides, for these two classification tasks, *Digits* and *Office-Home* respectively contain 10 and 65 categories. Top-1 and Top-5 accuracy are adopted for these two tasks.

Implementation Details. In federated learning process, all participants adopt the same hyper-parameter setting (*i.e.*, $\lambda_{Col} = 0.0051$ like [91] and $\lambda_{Loc} = 1$). Models are trained using Adam optimizer [31] with batch size of 512 and learning rate as 0.001 in both collaborative updating and local updating for all approaches. In terms of data scale, in *Digits* task, *MNIST*, *USPS*, *SVHN* and *SYN* are assigned to four participants. The size of corresponding private data is set to 150, 80, 5000 and 1800 respectively. As for *Office-Home* task, each participant is individually assigned with *Art*, *Clipart*, *Product* and *Real World*, and the corresponding private data size is 1400, 2000, 2500, 2000. The number of unlabeled public data is 5000 for these two tasks. For pre-processing, we resize all input images into 32×32 with three channels for compatibility. We do communication for $T = 40$ rounds, where all approaches have little or no accuracy gain with more communications rounds. Besides, for *SOLO*, models are trained on private data for 50 epochs, which are also initial models for federated learning process.

4.1. Comparison with State-of-the-Art Methods

We provide comparison results with state-of-the-art methods on two image classification tasks (*i.e.*, *Digits* and *Office-Home*) with three public data (*i.e.*, *Cifar-100*, *ImageNet* and *Fashion-MNIST*).

Inter Domain Analysis. We report the inter domain performance with state-of-the-art methods on Tab. 1. It clearly depict that under domain shift, *SOLO* present worst in these two tasks, demonstrating the benefits of federated learning. We observe that *FCCL* significantly outperforms better than counterparts. The Fig. 5 presents that *FCCL* achieves similar logits output between participants and redundancy within the logits output, confirming that *FCCL* successfully enforces the correlation of same dimensions and decorrelation of different dimensions on both public and private data.

Intra Domain Analysis. To compare the effectiveness of alleviating catastrophic forgetting, we show the intra domain performance in Tab. 2. Take the results of *Digits* task with *Cifar-100* as an example, our method outperforms the strong compared method, *RCFL*, by 2.30%. Besides, the intra domain accuracy via increasing communication rounds in Fig. 6a and optimization objective value in Fig. 6b present that *FCCL* suffers less periodic performance shock and is not prone to overfitting to current data distribution ($\bar{L}^{Loc} = 0.0225$), illustrating that *FCCL* is cable of balancing multiple knowledge, alleviating catastrophic forgetting.

Model Homogeneity Analysis. We further compare *FCCL* with other methods under model homogeneity. We set the shared model as *ResNet-18* and add the averaging parameters operation between *collaborative updating* and *local updating*. The Tab. 3 presents both inter and intra domain performance on *Office-Home* task with *Cifar-100*.

Methods	Digits				Office-Home			
	M	U	SV	SY	A	C	P	R
SOLO	70.20	74.19	74.57	73.60	65.27	60.50	74.68	54.28
FedMD [38]	77.30	80.05	77.73	87.72	66.17	60.63	76.35	56.60
FML [72]	80.66	79.75	78.58	88.87	81.46	65.58	79.82	65.07
RCFL [16]	82.59	81.05	78.79	91.40	65.13	61.33	76.44	55.78
FedDF [50]	82.95	78.84	78.46	91.30	66.10	60.44	75.70	55.98
FedMatch [28]	82.69	78.31	79.79	89.23	81.50	65.40	79.81	65.06
FCCL	88.84	84.42	78.55	91.23	81.51	65.42	79.84	65.16

Table 2. Comparison of intra domain performance with state-of-the-art methods on these two tasks with *Cifar-100*. The metric is evaluated on respective testing data in Eq. (8).

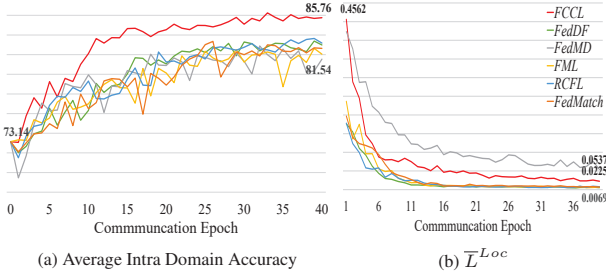


Figure 6. Comparison of intra domain performance and optimization objective value in local updating via increasing communication rounds on *Digits* task with *Cifar-100*.

Methods	Inter Domain				Intra Domain			
	A $\rightarrow$	C $\rightarrow$	P $\rightarrow$	R $\rightarrow$	A	C	P	R
SOLO	18.89	22.58	22.33	27.26	65.27	61.51	74.84	57.65
FedAVG [58]	57.85	54.05	55.72	60.18	66.71	60.90	74.29	57.49
FedMD [38]	61.03	62.41	62.45	62.55	66.50	61.75	73.63	58.10
FML [72]	39.56	36.94	32.73	42.00	74.87	60.73	77.19	60.71
RCFL [16]	61.52	59.56	57.56	63.59	67.16	61.39	73.33	58.58
FedDF [50]	61.10	57.92	62.19	60.41	66.69	60.69	74.12	57.69
FedMatch [28]	51.60	47.77	42.33	55.35	80.35	65.05	78.99	64.55
FCCL	64.48	62.33	63.26	64.86	81.38	65.47	79.40	65.19

Table 3. Comparison with state-of-the-art methods under model homogeneity on *Office-Home* task with *Cifar-100*.

4.2. Diagnostic Experiments

To demonstrate how each component in *FCCL* contributes to overall performance, a series of ablation experiments are conducted. The proposed method, *FCCL* is comprised of two components: *Federated Cross-Correlation Learning* and *Federated Continual learning*.

Federated Cross-Correlation Learning. To prove its robustness and stability, we evaluate the performance on different public data without label (*i.e.*, *Cifar-100*, *ImageNet* and *Fashion-MNIST*). The results in Fig. 7 suggest that *Federated Cross-Correlation Learning* achieves consistent performance in each domain. Moreover, it can be seen that it is more effective by the use of public data with rich categories (*ImageNet*) or simple detail (*Fashion-MNIST*).

Federated Continual Learning. We investigate the effectiveness of our core idea in handling catastrophic forgetting. As shown in Fig. 8, additionally considering dual-domain knowledge distillation (§ 3.2) leads to a substan-

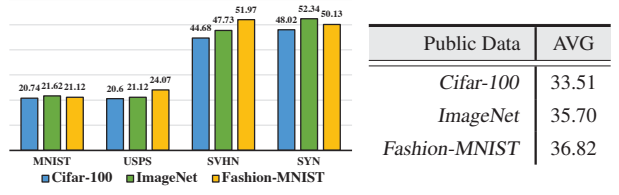


Figure 7. Ablation study on *Federated Cross-Correlation Learning* § 3.1 with different public data for inter domain performance on each domain performance (left) and overall performance (right) in *Digits* task.

tial inter domain performance gain (*i.e.*, 6.38% on *Digits* task with *Cifar-100*), compared with *w/o CON* (optimization objective in local updating is only cross-entropy loss, L_i^{Ce} in Eq. (3)). In addition, the Fig. 8 illustrates that it also boosts the intra performance (*i.e.*, 3.88% with *ImageNet*). The Fig. 4 visualizes features in intra and inter domain cases. As seen, the proposed *Federated Continual Learning* begets a well discriminative feature space. This suggests that exploiting extra restriction signals in local updating is beneficial for alleviating catastrophic forgetting.

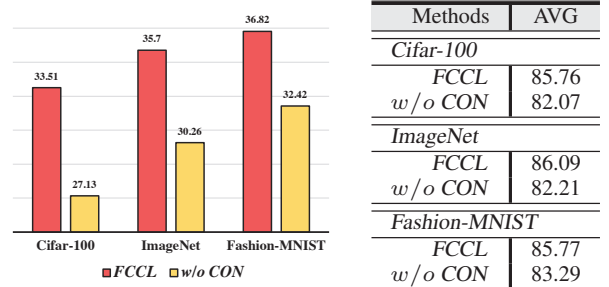


Figure 8. Ablation study on *Federated Continual Learning* § 3.2 for inter (left) and intra (right) domain performance on *Digits* task. *w/o CON* means that loss function is L_i^{Ce} in Eq. (3).

5. Conclusion

This paper proposes a simple and effective method of *FCCL* for federated learning. *FCCL* is capable of handling heterogeneity problem and alleviating catastrophic forgetting. In particular, we construct cross-correlation matrix in collaborative updating to learn a generalizable representation. Meanwhile, we introduce knowledge distillation with inter and intra domain information in local updating, boosting inter and intra domain performance. Experimental results on classification tasks show that our method performs favorably in comparison with state-of-the-art methods.

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